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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree 7th semester (S,FE) Exam April 2025 (2019 Scheme)

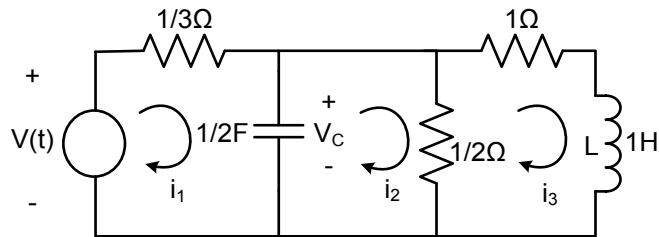
Course Code: EET401**Course Name: ADVANCED CONTROL SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| 1 | How do you choose the state variables for state space modelling? | (3) |
| 2 | Obtain the state model of the system described by the transfer function given as | (3) |
| | $\frac{Y(s)}{U(s)} = \frac{10(S + 4)}{S(S + 1)(S + 3)}$ | |
| 3 | State and prove any three properties of state transition matrix. | (3) |
| 4 | Derive the solution of homogeneous state equation | (3) |
| 5 | Consider the system described by the state model, $\dot{x} = Ax + Bu$ and $y = Cx$ | (3) |
| | A state feedback controller is employed with control law $u = -kx$ where k is the state feedback controller gain matrix. Obtain the modified state model of the system and give the block diagram. | |
| 6 | Discuss the procedure to test the controllability and observability using the PBH method. | (3) |
| 7 | Give the stability conditions based on describing function analysis. | (3) |
| 8 | How the non-linearities are classified? Give examples | (3) |
| 9 | Explain the terms phase plane, phase trajectory and phase portrait | (3) |
| 10 | Check the sign definiteness of the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 0 \end{bmatrix}$ | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

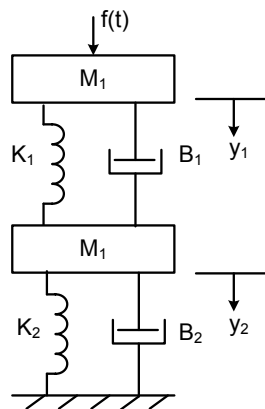
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|----|---|-----|
| 11 | a) Derive the state model of the electrical circuit shown below | (9) |
|----|---|-----|



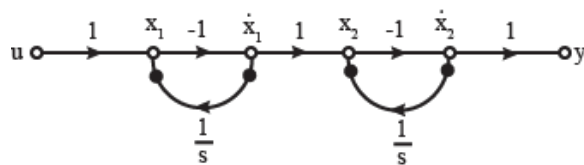
- b) Using the similarity transformation, diagonalize the matrix $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ (5)

OR

- 12 a) Derive the state model of the mechanical system shown below (9)



- b) Determine the state model of the system shown in the signal flow graph given below (5)



Module II

- 13 a) Obtain the unit step response of the system describe by the state model given below for the initial condition, $x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ (8)

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \ 0] x$$

- b) Compute the state transition matrix using Cayley-Hamilton theorem for the system with system matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$ (6)

OR

- 14 a) Determine the transfer function of the system describe by the state model given below (8)

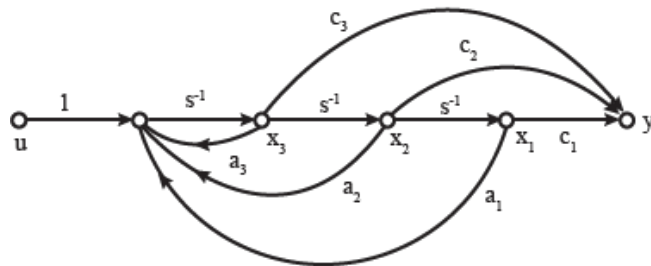
$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = [0 \quad 0 \quad 1]x$$

- b) Explain Cayley-Hamilton theorem with an example (6)

Module III

- 15 a) Consider the state space system expressed by the signal flow diagram shown in the figure. Test the controllability using Kalman's method (7)



- b) A regulator has the plant described in the state model given below (7)

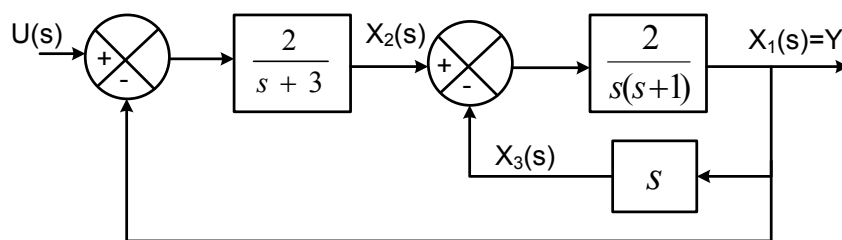
$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 20.6 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0]x$$

Design a state feedback controller with the control law $u = -KX$ such that the closed loop system has the eigenvalues at $-1.8 \pm j2.4$

OR

- 16 a) Test the controllability and observability of the system shown in the block diagram given below (8)



- b) Design a full order state observer for the system described in the state model given below (6)

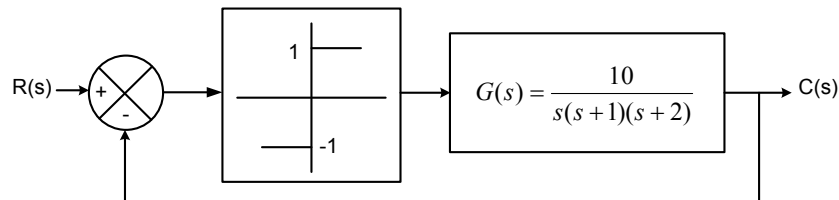
$$\dot{x} = \begin{bmatrix} 0 & 20.6 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [0 \quad 1]x$$

The desired eigenvalues of the observer matrix are $\mu_1 = -10$ and $\mu_2 = -10$

Module IV

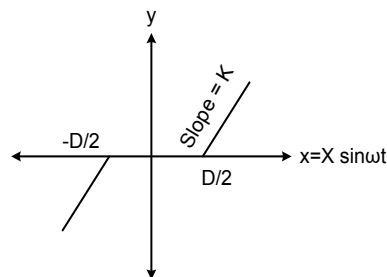
- 17 a) Determine the frequency and nature of the limit cycle for the unity feedback system given below. (10)



- b) State the assumptions made in the describing function method of analysis (4)

OR

- 18 a) Derive the describing function of the non-linearity shown below (8)



- b) Explain the following i) Jump resonance ii) Limit cycle iii) Frequency entrainment (6)

Module V

- 19 a) Explain the stable and unstable limit cycles using necessary sketches. (4)

- b) A linear system is describe by the state equation $\dot{x} = Ax$ where $A = \begin{bmatrix} -4K & 4K \\ 2K & -6K \end{bmatrix}$ (10)

Using the Lyapunov method, find the restrictions on the parameter K to guarantee the system stability.

OR

- 20 a) Determine the singular points of the system described by the differential equation (6)

$$\ddot{x} + 4\dot{x} - 2.5x^2 + 5x = 0$$

- b) Determine the stability of the system described by the state equation given below (8)
using Lyapunov method. Select the Lyapunov function as

$$V(x) = x_1^4 + (x_1 + x_2)^2$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_2 - x_1^3$$
